

# PRODUCT REQUIREMENTS DOCUMENT

## Twilio Platform — twilio.com

Version: 2.0 | Date: September 2026 | Prepared for: Hi Tech Agentic Test Pipeline

Start URL: <https://twilio.com>

## 0. Global Behavioral Policies

The following policies apply to every user flow in this document. They describe how the agent should behave during test execution, what kind of interactions are expected, and how to handle environmental interruptions.

### 0.1 Overlay Handling

Twilio.com deploys overlapping interstitial elements: a cookie-consent banner, a live chat widget, promotional modals, and GDPR consent dialogs. Before every interaction step, the agent must dismiss any visible overlay by clicking its close control or pressing Escape, then wait for the element to be removed from the DOM.

### 0.2 Interaction Model

Tests in this document follow an **action** → **observe** → **verify** model. Each acceptance criterion specifies: (a) the navigation path, (b) the UI elements to interact with, (c) the input data where applicable — all self-contained so no external credentials or account details are ever required — and (d) the observable outcome to assert.

### 0.3 Self-Contained Test Data

Where a flow involves typing into a form field, the exact value to enter is specified inline within the acceptance criterion (e.g. *test.user@example.com*, *sms*, *+15005550006*). The agent uses these values verbatim and does not pause to look up, generate, or request any additional information. If a page transitions to a step that asks for information not provided in the criterion, the agent stops at that boundary and records the transition as the observed outcome.

### 0.4 Boundary Conditions

Flows that reach a sign-in wall, email-verification gate, or payment screen treat the appearance of that gate as the final observable outcome to assert — they do not attempt to proceed past it. The acceptance criterion for such flows explicitly names the expected gate element so the agent knows when to stop and record a pass.

## 1. Overview

Twilio is a cloud communications platform offering programmable Voice, SMS, Email, and Video APIs. This PRD defines 5 user flows comprising 10 acceptance criteria for the Kane-CLI agentic pipeline. Each flow exercises a meaningful user journey through publicly accessible pages, interacting with navigation, content sections, search inputs, and form fields using self-contained test data.

## 2. Scope — 5 Flows, 10 Acceptance Criteria

- Flow 1: Homepage journey — scroll feature section, click product CTA
- Flow 2: Docs search — type a query, verify live results appear

- Flow 3: Pricing interaction — toggle billing cycle, verify price update
- Flow 4: Products deep-dive — navigate to product page, reach code example
- Flow 5: Login form validation — submit invalid data, verify error response

## 3. Functional Requirements

### 3.1 Homepage Journey

| ■ INTERACTION DEPTH NOTE                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                         |                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Each acceptance criterion in this section describes a complete user interaction journey — not just a page load check. The agent should perform each described action (click, type, scroll, hover) in sequence and verify the observable outcome at each step. All test data used in form fields is self-contained within each criterion and requires no external lookup or account-specific information.                    |                                                                                                                         |                                                                                                                                                                                                                                                                                                                                      |
| ■ OVERLAY HANDLING — apply before every interaction in this flow                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                         |                                                                                                                                                                                                                                                                                                                                      |
| Before clicking any element, scan the viewport for cookie banners, chat widgets, promotional modals, or GDPR consent dialogs. Dismiss each by clicking its labelled close control ('Accept All', 'No thanks', 'X', 'Got it') or pressing Escape. Wait for the element to disappear from the DOM before proceeding. If a widget re-opens, dismiss it again. If dismissal fails after 3 attempts, log a warning and continue. |                                                                                                                         |                                                                                                                                                                                                                                                                                                                                      |
| REQ ID                                                                                                                                                                                                                                                                                                                                                                                                                      | Requirement                                                                                                             | Acceptance Criteria                                                                                                                                                                                                                                                                                                                  |
| HM-01                                                                                                                                                                                                                                                                                                                                                                                                                       | Homepage renders hero section and user can scroll to the product feature cards below the fold                           | Agent navigates to <a href="https://twilio.com">https://twilio.com</a> ; hero headline is visible; agent scrolls down until the product feature/card section comes into view (e.g. 'Communications APIs', 'Engage customers'); at least 3 feature cards or product tiles are visible in the viewport after scrolling.                |
| HM-02                                                                                                                                                                                                                                                                                                                                                                                                                       | Primary CTA click leads to a reachable next-step page (sign-up gate or demo request form) without a 404 or blank screen | Agent clicks the primary CTA button on the homepage ('Start for free' or 'Get a demo' — whichever is present); the resulting page URL changes and a heading or form element appears on the new page; if a sign-up wall loads, the presence of an email input field is recorded as the expected boundary outcome and the test passes. |

### 3.2 Documentation Search Journey

| ■ INTERACTION DEPTH NOTE                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
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| ■ OVERLAY HANDLING — apply before every interaction in this flow                                                                                                                                                                                                                                                                                                                                                            |  |  |
| Before clicking any element, scan the viewport for cookie banners, chat widgets, promotional modals, or GDPR consent dialogs. Dismiss each by clicking its labelled close control ('Accept All', 'No thanks', 'X', 'Got it') or pressing Escape. Wait for the element to disappear from the DOM before proceeding. If a widget re-opens, dismiss it again. If dismissal fails after 3 attempts, log a warning and continue. |  |  |

| REQ ID | Requirement                                                           | Acceptance Criteria                                                                                                                                                                                                                                                                                       |
|--------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DC-01  | Docs landing page is reachable and renders a functional search input  | Agent navigates to <a href="https://www.twilio.com/docs">https://www.twilio.com/docs</a> ; the page loads without error; a search input field (placeholder text 'Search', 'Search docs', or similar) is visible and accepts keyboard focus when clicked.                                                  |
| DC-02  | Typing a product keyword into the docs search returns visible results | Agent clicks the search input on the docs page and types the value <b>sms</b> ; within 5 seconds a results list, dropdown panel, or filtered page section appears containing at least 1 entry whose text includes 'SMS', 'Messaging', or 'Programmable'; the results are visible without requiring login. |

### 3.3 Pricing Page Interaction Journey

| ■ INTERACTION DEPTH NOTE                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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| ■ OVERLAY HANDLING — apply before every interaction in this flow                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                   |
| Before clicking any element, scan the viewport for cookie banners, chat widgets, promotional modals, or GDPR consent dialogs. Dismiss each by clicking its labelled close control ('Accept All', 'No thanks', 'X', 'Got it') or pressing Escape. Wait for the element to disappear from the DOM before proceeding. If a widget re-opens, dismiss it again. If dismissal fails after 3 attempts, log a warning and continue. |                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                   |
| REQ ID                                                                                                                                                                                                                                                                                                                                                                                                                      | Requirement                                                                                                            | Acceptance Criteria                                                                                                                                                                                                                                                                                                                                                               |
| PX-01                                                                                                                                                                                                                                                                                                                                                                                                                       | Pricing page loads and displays at least 2 product pricing blocks each containing a product name and a price indicator | Agent clicks 'Pricing' in the top nav; URL contains '/pricing'; at least 2 pricing blocks are visible, each showing a product name (e.g. 'Messaging', 'Voice') and a price label (e.g. '\$0.0075', 'Pay as you go', 'Starting at \$X').                                                                                                                                           |
| PX-02                                                                                                                                                                                                                                                                                                                                                                                                                       | Billing cycle toggle (monthly / annual) updates displayed prices or labels when clicked                                | Agent locates a billing period toggle or tab on the pricing page (e.g. 'Monthly' / 'Annually'); clicks the non-default option; verifies that at least one price label or CTA text on the page changes or that a discount badge appears — confirming the toggle is interactive and drives visible UI change. If no toggle exists, agent records 'single pricing model' and passes. |

### 3.4 Product Deep-Dive Journey

| ■ INTERACTION DEPTH NOTE                                                                                                                                                                                                                                                                                                                                                                                 |  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| Each acceptance criterion in this section describes a complete user interaction journey — not just a page load check. The agent should perform each described action (click, type, scroll, hover) in sequence and verify the observable outcome at each step. All test data used in form fields is self-contained within each criterion and requires no external lookup or account-specific information. |  |  |

#### ■ OVERLAY HANDLING — apply before every interaction in this flow

Before clicking any element, scan the viewport for cookie banners, chat widgets, promotional modals, or GDPR consent dialogs. Dismiss each by clicking its labelled close control ('Accept All', 'No thanks', 'X', 'Got it') or pressing Escape. Wait for the element to disappear from the DOM before proceeding. If a widget re-opens, dismiss it again. If dismissal fails after 3 attempts, log a warning and continue.

| REQ ID | Requirement                                                                                                        | Acceptance Criteria                                                                                                                                                                                                                                                                                                    |
|--------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PR-01  | Products navigation menu opens and lists product categories the user can read without interacting further          | Agent hovers over or clicks 'Products' in the top nav; a mega-menu or dropdown opens; at least 3 labelled product categories are visible (e.g. 'Messaging', 'Voice & Video', 'Email', 'Segment', 'Flex'); the menu remains open long enough to be observed.                                                            |
| PR-02  | Clicking a product link in the menu loads a product detail page that contains a code sample or API snippet section | Agent clicks the first product link visible in the Products menu (e.g. 'Programmable Messaging' or 'SMS'); the product detail page loads; agent scrolls down until a code block, API snippet, or 'Sample code' / 'How it works' section is visible in the viewport; the code block contains at least one line of text. |

### 3.5 Login Form Validation Journey

#### ■ INTERACTION DEPTH NOTE

Each acceptance criterion in this section describes a complete user interaction journey — not just a page load check. The agent should perform each described action (click, type, scroll, hover) in sequence and verify the observable outcome at each step. All test data used in form fields is self-contained within each criterion and requires no external lookup or account-specific information.

#### ■ OVERLAY HANDLING — apply before every interaction in this flow

Before clicking any element, scan the viewport for cookie banners, chat widgets, promotional modals, or GDPR consent dialogs. Dismiss each by clicking its labelled close control ('Accept All', 'No thanks', 'X', 'Got it') or pressing Escape. Wait for the element to disappear from the DOM before proceeding. If a widget re-opens, dismiss it again. If dismissal fails after 3 attempts, log a warning and continue.

| REQ ID | Requirement                                                                                                                                                                   | Acceptance Criteria                                                                                                                                                                                                                                                                                                                                                                          |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LG-01  | Login page is reachable from the top navigation and renders email and password input fields                                                                                   | Agent clicks 'Log in' in the top nav; the browser navigates to the Twilio Console login page (login.twilio.com or console.twilio.com); both an email/username input field and a password input field are present in the viewport.                                                                                                                                                            |
| LG-02  | Submitting the login form with the value <i>nouser@example.com</i> as email and <i>WrongPass123!</i> as password produces a visible inline error message without a page crash | Agent clicks the email field and types <b>nouser@example.com</b> ; clicks the password field and types <b>WrongPass123!</b> ; clicks the 'Log in' or 'Sign in' submit button; within 8 seconds an inline error message appears on the page (e.g. 'Incorrect email or password', 'Invalid credentials', or equivalent); the page does not white-screen, crash, or navigate away unexpectedly. |

## 4. Test Scenarios (10 — maximum)

Each scenario maps to one acceptance criterion and describes the full user journey the agent must execute. All input values are embedded in the scenario description — the agent does not need to source any data from outside this document.

| #  | Scenario (full journey)                                                 | Req   | Pri |
|----|-------------------------------------------------------------------------|-------|-----|
| 1  | Homepage: load page → scroll to feature cards → assert 3+ cards visible | HM-01 | P1  |
| 2  | Homepage: click primary CTA → assert new page or sign-up gate loads     | HM-02 | P1  |
| 3  | Docs: navigate to /docs → assert search input is focusable              | DC-01 | P1  |
| 4  | Docs: type 'sms' in search → assert results dropdown/panel appears      | DC-02 | P1  |
| 5  | Pricing: open /pricing → assert 2+ product blocks with price labels     | PX-01 | P1  |
| 6  | Pricing: click billing toggle → assert price or label changes on page   | PX-02 | P2  |
| 7  | Products: hover Products menu → assert 3+ category labels visible       | PR-01 | P1  |
| 8  | Products: click first product link → scroll to code sample section      | PR-02 | P2  |
| 9  | Login: navigate to login page → assert email + password fields present  | LG-01 | P1  |
| 10 | Login: enter nouser@example.com + WrongPass123! → assert error message  | LG-02 | P1  |

## 5. Pipeline Configuration

- **requirements\_doc:** twilio\_PRD.pdf
- **start\_url:** https://twilio.com
- **test\_language:** javascript
- **max\_tests\_to\_run:** 10

### Required Secrets

- KANE\_USERNAME / KANE\_ACCESS\_KEY — TestMu credentials
- LT\_USERNAME / LT\_ACCESS\_KEY — LambdaTest credentials

### Known Overlay Selectors on twilio.com

The following CSS selectors identify the most common overlays on twilio.com. The agent should attempt to dismiss elements matching these selectors before each interaction step:

- **#onetrust-accept-btn-handler** — cookie consent accept button
- **.truste\_overlay, .truste\_box\_center** — TRUSTe consent box
- **[aria-label='Close'], [aria-label='Dismiss']** — generic modal close
- **.drift-frame-controller** — Drift live chat widget
- **button:contains('No thanks'), button:contains('Maybe later')** — promotional modals